Dual Role on Shoulders: Who Benefits from Platform Exclusivity?

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Abstract

A digital platform can play a dual role after vertical integration: running marketplaces for third-party products, while selling their own products on those marketplaces. This paper examines a platform' exclusivity decision and its welfare impact with this controversial practice. Using a model with Nash bargaining and Bertrand competition, I show that the platforms may be willing to give up their exclusivity. However, risk of collusion and potential anti-competitive impact may also arise from this practice.

1 Introduction

Two-sided platforms bring two groups who use the platform, as with buyers and sellers of some underlying product. For example, video game consoles such as the Sony PlayStation and the Microsoft Xbox provide the platform on which consumers play video games provided by content developers. In such markets, products may or may not be exclusive. For example, from its release in 2018 until 2022, the popular game "God of War" (developed by Santa Monica Studio) was available only on the PlayStation. In contrast, the very popular Fortnite game (developed by Epic Games) has long been available on the PlayStation, the Xbox and other platforms.

In recent years platform providers in a variety of markets have opted for a vertical integration strategy in which they produce content for sale on the platform. For example, video platforms such as Amazon Prime Video, Apple TV, and Netflix all produce much of the video content they show. In the video game market, both Microsoft and Sony have acquired multiple previously independent game developers. One possible explanation for such vertical integration is that exclusivity might be easier to achieve and maintain. However, integrated platforms do not always opt for exclusivity and sometimes make their products available on other platforms, as in the video game market.

The primary research question of this paper concerns when integrated platforms will choose to sell their products on an exclusive basis and when they will choose to "cross-sell"-also selling on another platform. In addition, I examine the comparative welfare properties (as measured by consumer and producer surplus) of exclusive and cross-selling.

To answer this question, I build a tractable duopoly model with two differentiated contents and two platforms integrated with content providers who can adopt the cross-selling strategy by giving up their exclusivity and selling on the rival's platform. An overall 3-stage game is played by the platforms. In the first stage, each firm decides whether to cross-sell. The second stage is a Nash bargaining stage in which, if cross-selling occurs, the firms bargain over the commission. The third stage is a Bertrand pricing stage. While the model is sufficiently general to apply to many platform markets, it is motivated by the video game market.

I study two cases. First, when consumers can't move across platforms(singlehoming), if one platform, say platform A, sells exclusively, consumers on the other platform don't have access to platform A's content. Second, when consumers can move across platforms(multihoming), consumers can get access to exclusive content on other platforms with an additional constant console fee. In each of the cases, there are three potential equilibriums: i). an exclusive equilibrium where both firms sell exclusively, ii). a full-cross-selling equilibrium where both firms cross-sell, and iii). a semi-cross-selling equilibrium where one firm cross-sells and the other sells exclusively.

In traditional view of antitrust, one might think that vertical integration brings input foreclosure, and thus makes exclusivity more likely to occur. However, our results suggest that in both cases, the optimal first-stage strategy for both firms is give up exclusivity and supply their rival, thus the overall 3-stage game will end up in a full-cross-selling equilibrium. The main driver of this strategy is not the extra revenue generated from selling on another platform as in this equilibrium both firms would agree on a 100% commission, leaving none revenue for the product owner. I explain this result in the perspective of collusion. After a vertical integration, the trade-off of product exclusivity changed dramatically for both the platforms and the content providers. A bargain between a platform and a content providers is transformed into a profit-split between platforms, thereby generating incentives for the platforms to push up the market price together with the commission as a tool.

This non-exclusivity can potentially have anti-competitive effect. When the console fee is lower than its upper limit, the full-cross-selling equilibrium generates a consumer surplus higher than the singlehoming exclusive equilibrium but lower than the multihoming exclusive equilibrium. One potential explanation is that the exclusive equilibrium under singlehoming

works as a monopoly market while it works as a competitive market under multihoming. As a result of the higher price generated by the 100% commission rate, consumers' welfare is harmed.

As for other equilibrium characteristics, under singlehoming, firms' equilibrium behaviors depend only on the degree of differentiation between the two products. As the substitutability increases, the negotiated commission increases and approaches 1 as the products approach being perfect substitutes. In full-cross-selling equilibrium and semi-cross-selling equilibrium increased substitutability reduces profits. Under multihoming, firms' behaviors depend not only on the degree of content differentiation but also on the console fee. The negotiated commission would be larger as the substitutability and the console fee grow larger. The equilibrium profits in all 3 cases decrease with substitutability

I further extend our analysis by replacing the demand system with general functions and several essential assumptions and proved that most of our findings still hold with these alternative demands.

This paper is organized as follows. Section 2 provides a review of related literature and of the industry that motivates the model. Section 3 sets out our baseline model, explaining how the game is played, how consumers react to the market, and how the firms interact with each other under different strategies. Sections 4-6 solves the 3 stages of the game respectively. Section 7 explains the competitive impact with a general form demand system and Section 7 concludes.

2 Background

2.1 Related Literature

Our research studies how platforms' profits and consumer welfare are impacted by their exclusive strategies when they are vertically integrated with content providers. This paper is related to several streams of industrial organization literature, yet, above all, it's motivated by the seminal papers on "two-sided market."(Rochet and Tirole, 2006; Caillaud and Jullien, 2003)

Our paper is related to the large theoretical literature on market exclusivity. Though it has been modeled in various ways, exclusive contracts have been proved to be a powerful tool for the platforms to manage homing decisions on the two sides of the market. How the platforms and content providers divide the overall profit plays a crucial role in determining the equilibrium output. Armstrong (1999) provides one of the first studies focusing on the exclusive contract in media platforms. With independent content providers and platforms financed only through subscription fees, it has been found that a lump-sum payment for content pushes for exclusive

contracts more than a per-subscriber fee. Similarly, Harbord and Ottaviani(2001)'s research shows that content providers find it most profitable to sell exclusively for lump-sum payment while it's optimal for the platform to use a per-subscriber fee. Hagiu and Lee(2011) build a model with two platforms and a continuum of content providers, where platforms can offer content providers transfers contingent on whether they join exclusively or not. They specify two ways how content providers and the platforms set the price and divide the revenue based on who controls pricing. Their equilibrium result shows that content providers' exclusive decisions ultimately depend on who controls the retail price: if the content providers totally sell their content and give up control, they tend to sell exclusively.

On top of the exclusivity research, another stream of literature studies how vertical integration can change exclusive decisions for the platforms and content providers. Weeds(2016) shows that vertically integrated platforms have an incentive to withhold premium content from their rivals if only there exists some form of scale effect such that the asymmetry induced by content exclusivity raises industry profits. On the contrary, D'Annunzio(2017) finds that the premium content will always be provided exclusively no matter whether there is vertical integration or not, yet vertically integrated platforms have more incentives to invest in the quality of their exclusive content.

Carroni et al(2018) focus on the Superstar products, and find that exclusivity is less likely to occur under vertical integration than under vertical separation. Their explanation lies in the trade-off of vertical integration: after acquiring the product, the loss of the platform from giving up their exclusivity is small compared with the gain from larger demand when selling their product on their rival's platform.

While rich theoretical research tries to find how firms would act in equilibrium, empirical papers focus more on the competitive impact of vertical integration. Foreclosure is one major reason why vertical integration is thought anti-competitive, but the empirical evidence is still far from definitive. Ali & Chad(2007) estimated its impact using a rich data set of cement and ready-mixed concrete plants. Their result shows that with a greater presence of vertically integrated firms in a market comes lower prices and higher quantities, suggesting no anti-competitive foreclosure effect. Lee(2013) studies exclusivity in the industry of video games and concludes that exclusivity can increase sales and consumer welfare by favoring the entrant platforms.

Some researchers believe that vertical integration can also have pro-competitive effects. As summarized by Slade(2022), the dominating reason behind is the elimination of successive monopoly markups or double marginalization(EDM). Crawford et. Al. (2018) estimate a

structural model in the regional sports TV network market. They find a significant efficiency-enhancing effect of vertical integration by reducing double marginalization and increasing carriage of channels, with potential welfare harm due to foreclosure. However, as Salinger(1991) shows in a theoretical work, EDM may not be efficient when the integrated firms produce multiple products. Luco and Marshall(2020) show evidence in the carbonated-beverage industry that when Coca-Cola Company and PepsiCo acquired some of their bottlers, the price of their own price decreased. However, the price of their integrated bottlers' products increased, making the overall competitive effect ambiguous.

Our research is also related to a new trend of literature that studies platforms' preference over the roles under different conditions and their impact on platform competition. Hagiu and Wright(2015) recognize that platforms can choose between functioning as a marketplace or as a reseller, whose fundamental distinction lies in the allocation of control rights over noncontractible decisions, such as prices, advertising, etc. The optimal mode in their mode is driven by market characteristics such as the relative importance of suppliers and private information, spillover effects of marketing activities, and product features. Hagiu et al(2022) further show that the dual mode (acting as both marketplace and seller) is always better than any one of the single roles for the platform. Moreover, a ban on the dual mode will harm consumers and decrease overall welfare.

2.2 The Video Game Industry

With the revolutionary development in graphics programming, video games have evolved from mechanical arcade cabinets to modern home consoles, from pixelated figures to cinematic sceneries, from a coin-based domain of children and hobbyists to a $269 billion worldwide comprehensive industry consisting of software, hardware, social media, e-sport, etc. Nowadays, 67% of American adults recognize themselves as gamers, and 75% of American s have at least one video game console in their household(Tripp et al., 2020).

The specific video game type I am analyzing in this paper is the console game. A typical console game system comprises a hardware platform(the console) and software (the games). Each platform works as a two-sided market with one side of game developers and one side of gamers. Consumers need to pay for the console and each game they want to play.

Nowadays, Nintendo, Sony, and Microsoft take up the whole console market. They are allowed and are enthusiastic about organizing their own game studios and acquiring game developers. Any titles produced or owned by the platform are called a first-party title, while the others are called third-party titles that need to pay a royalty for every game sold. Most

likely, since the consoles have little if not none stand-alone utilities for consumers, the platforms will invest heavily in their first-party titles and make them exclusive to attract gamers. At the same time, independent game developers normally don't have comparable size as the first-party developers, leading to the most top-selling game, or the Superstar games, being first-party exclusive games. On Sony's PlayStation 4, one of the three giant platforms of game consoles, 8 of 10 top-selling games are first-party exclusive games, while only 1 of the top 20 most-selling games on Nintendo Switch is a third-party title.

Recently, there was a surge of acquisitions in the console game industry. All the 10 largest mergers and acquisitions in this industry took place in the last 6 years, 8 of which happened in the last 3 years. In Sept 2020, Microsoft acquired a video game studio named ZeniMax with $8.1 billion, which was the all-time high record for video game studio acquisition until 16 months later, when they spent another $68.7 billion on the acquisition of Activision Blizzard. At about the same time, Sony acquired 10 independent game studios in the last 3 years, with the highest value of $3.7 billion. However, when most analysts believe that the two giants are allocating resources for an exclusivity war, they both made claims that all the games from the lately-acquired studios will remain available on all the other platforms.

3 Model Preliminaries

In this section, I consider a simplified model of the video game industry. Specifically, I consider a market with only two platforms i = 1 , 2, in which consumers can only buy the contents they provide.

Consumers. On each of the platform, there exist a group of prior consumers with identical preference. In most of the previous studies on two-sided market, consumers immediately enjoy the utility of the platform itself and all the contents provided when they join the platform. I think it's not ideal for a lot of nowadays content platforms for two main reasons. First of all, a hardware platform can hardly generate standalone utility for consumers. Imagine a Kindle without an ebook, an iPad without software, and a PlayStation 5 without a game. They are simply useless for the consumers without content. Secondly, consumers can not enjoy most of the attractive contents before they purchase them. Even for some free software and games, unlocking some essential functions requires payment.

Therefore, I modified consumers' utility generation process. They don't receive any standalone utility from joining the platform nor from any fringe game, but from purchasing the 2 differentiated superstar games on the market, i.e., game x and game y. This captures that in practice when consumers decide which platform to join, they primarily focus on the superstar

products, and the fringe products play the role of extra bonus. Each consumer can purchase any game available on his/her platform, but can buy at most one of each game.

If both games are available on platform i, the demand for each of the products on each of the platforms can be derived from the following symmetric quadratic utility function:

$$U_i = a\left(x_i + y_i\right) - \frac{1}{2}\left(x_i^2 + y_i^2\right) - sxy \tag{1}$$

Where a is a large enough exogenous parameter to ensure positive demand, represents the demand of game x on platform i, s stands for the degree of substitutability between. Here I assume the games to be substitutes, so s must be positive and cannot exceed 1, i.e, $0 < s < 1$. Note this utility function implicitly reflects the preference of many consumers as presumably each consumer buys at most one unit of each game. Let Px and Py denote the price of each game, we can derive the demand functions to be:

$$P^X = a - x - sy \tag{2}$$

$$P^Y = a - y - sx \tag{3}$$

Platforms. The two platforms are vertically integrated with one of the superstar games. Specifically, platform 1 owns game x, and platform 2 owns game y. Platforms can only set the price for their own game, and it will be sold at that price on any platform.

Because of the dual role of a vertically integrated platform, i.e., the seller and the marketplace, their revenue consist of two parts: the revenue generated by selling their own product and by selling other firms' product. In most real-world platform businesses, the platforms can earn money from each of the transactions that take place on it by a fixed amount or proportion. In our model, since this research primarily focus on the "superstar" products that can attract subscriptions, it is allowed that they have bargain power.

To be more specific, if one game is sold on the rival's platform, the two platforms can negotiate and determine a commission rate $(1 - \sigma)$ owing to the platform, and the owner of the product earns the rest. For instance, if platform 1 adopts the cross-selling strategy while platform 2 doesn't, the revenues of the two platforms can be described as the following:

$$\Pi_1 = P^X x_1 + \sigma P^X x_2 \tag{4}$$

$$\Pi_2 = (1 - \sigma) P^X x_2 + P^Y y_2 \tag{5}$$

The selection of the commission can reflect the preference of the platforms over the dual role. If they end up selecting a large commission, then, relatively, they earn more by selling their

own products instead of other products, which suggests that the seller role is more appreciated by the platforms.

For most digital content, such as games, movies, and series, the majority of the cost occurs during production and development, and the cost of distributing one additional copy is trivial. Accordingly, the cost of the content is, to some extent, sunk for the firm since they cannot undo it over the period of a game analyzed in this paper. Therefore, I model the cost to be irrelevant to the platform's strategy and normalize it to be 0. Moreover, since the cost is 0, then the revenue should be the same as the profit.

Our overall equilibrium concept is a sequentially rational equilibrium in the following three-stage game:

1. Both platforms decide whether to adopt cross-selling.

2. The platforms engage in Nash bargaining over the commission unless no cross-selling occurred, nor multihoming allowed. If they settle, they play along with the determined commission. If they fail to settle, then cross-selling is prohibited in the following game.

3. Platforms play a Bertrand price game

4 Stage 3: The Bertrand Price Game

In this section, I solve the third stage of the overall game: the Bertrand price game. I will first analyze the case when consumers can't move across the platforms, i.e., the singlehoming case. This case can capture the real-world scenario when few consumers actually change their hardware platform due to high console fees or brand loyalty. For example, most consumers on Windows PC need to seriously consider the necessity of switching to MacOS due to the high price of a Macbook, and some Macbook users won't switch to Windows PC because of their loyalty to Apple and MacOS.

After the singlehoming case, I then allow consumers to move across platforms(the multihoming case). The crucial change is that now consumers can purchase the exclusive game on another platform by purchasing the hardware platform first. I will discuss in detail in later sections.

In both cases, there are three possible equilibriums: i) Exclusive Equilibrium: Both firm sells their game exclusively. ii) Full-Cross-Selling Equilibrium: Both firms sell their game on both platforms. iii) Semi-Cross-Selling Equilibrium: One platform gives up its exclusivity while the other one keeps it. WLOG, I will name the firm that cross-sells firm 1, and the other firm 2.

4.1 The Singlehoming Case

Exclusive Equilibrium

If neither of the platforms adopts cross-selling strategy, then each of the games would only be sold on their own platform. Thus consumers on platform 1 would not have access to game y and vice versa. The utilities of consumers would be simplified to the followings:

$$U_1 = ax_1 - \frac{1}{2}x_1^2$$
$$U_2 = ay_2 - \frac{1}{2}y_2^2$$

With budget constraints, we can derive the demands as follows:

$$x_1\left(P^X\right) = a - P^X$$
$$y_2\left(P^Y\right) = a - P^Y$$

For the platforms, when they both sell exclusively, they only make profits by selling on their own platform. Therefore, both platforms become a monopoly market. Using the demands, the profits of platforms can be represented by:

$$\Pi_1 = P^X\left(a - P^X\right)$$
$$\Pi_2 = P^Y\left(a - P^Y\right)$$

Each platform will set its price to optimize its profit. Solving the maximization problem, we can obtain the equilibrium price of game x and y. Since both the utility functions and the profit functions are symmetric, we end up one price for both game:

$$P_E^X = P_E^Y = \frac{a}{2} \tag{6}$$

And immediately, the equilibrium profits are:

$$\Pi_1 = \Pi_2 = \Pi_{NC} = \frac{a^2}{4} \tag{7}$$

Full-Cross-Selling Equilibrium

If both platforms adopt cross-selling strategy, then all the consumers have access to both games, and their utilities can be represented by:

$$U_i = a\left(x_i + y_i\right) - \frac{1}{2}\left(x_i^2 + y_i^2\right) - sxy$$

Note this case is actually a traditional Bertrand model with quadratic utility function. The demand can be derived as:

$$x_1 = x_2 = \frac{1}{1-s^2}\left[a\left(1-s\right) - P^X + sP^Y\right]$$
$$y_1 = y_2 = \frac{1}{1-s^2}\left[a\left(1-s\right) - P^Y + sP^X\right]$$

When both firms cross sell, each of them earns 3 parts of profits: i) Selling their own game on their own platform. ii) Selling their own game on rival's platform. iii) Selling rival's game on their own platform. In math, they can be represented as:

$$\Pi_1 = P^X x_1 + \sigma P^X x_2 + \left(1-\sigma\right) P^Y y_1$$
$$\Pi_2 = P^Y y_2 + \sigma P^Y y_1 + \left(1-\sigma\right) P^x x_2$$

Platform 1 and 2 would maximizing their profit by setting the optimal price for their games. Similar as the exclusive equilibrium, the firm's profit functions are symmetric, therefore, at equilibrium, we have one price for both game:

$$P_C^X = P_C^Y = \frac{\left(1-s\right)\left(1+\sigma\right)}{1+\sigma-s}\frac{a}{2} \tag{8}$$

Since both s and σ are ranged from 0 to 1,

$$\left(1-s\right)\left(1+\sigma\right) = 1+\sigma-s-s\sigma < 1+\sigma-s \tag{9}$$

We know both-cross-selling equilibrium generates a price that's always lower than the exclusive equilibrium.

Semi-Cross-Selling Equilibrium

WLOG, I will name the firm who cross sells firm 1, and the other one firm 2. Accordingly, both games will be available on platform 2, but only game 1 on platform 1. The utilities of their consumers can be expressed by:

$$U_1 = ax_1 - \frac{1}{2}x_1^2 - P^X x_1$$
$$U_2 = a\left(x_2+y_2\right) - \frac{1}{2}\left(x_2^2+y_2^2\right) - sx_2y_2$$

We can derive the demands to be:

$$x_1 = a - P^X$$
$$x_2 = \frac{1}{1-s^2}\left[a\left(1-s\right) - P^X + sP^Y\right]$$
$$y_2 = \frac{1}{1-s^2}\left[a\left(1-s\right) - P^Y + sP^X\right]$$

In this case, the two firms earns two different parts of profits. Firms 1 make profits by selling their own game on both platforms, while firm 2 makes profits by selling their game and firm 1's game on their own platform. In math, the profits of both platforms can be presented by:

$$\Pi_1 = P^X x_1 + \sigma P^X x_2$$
$$\Pi_2 = P^Y y_2 + (1-\sigma) P^x x_2$$

Since the demand of one game is always negatively related to its own price, each firm's profit is single-peaked in their own price. To solve the optimization problem, we can use the following first order conditions:

$$\partial \Pi_1 / \partial P^X = 0 \quad ; \quad \partial \Pi_2 / \partial P^Y = 0$$

Equilibrium Comparison

To better demonstrate our result, I did a simulation with $a = 40$ and $s = 0.6$.

Among the three cases, asymmetric equilibrium only occurs when there is one platform cross-selling and the other doesn't. Compared with exclusive equilibrium, all the other equilibriums generate lower prices, and accordingly higher consumer surplus. As σ grows larger, the seller role is more appreciated by the platforms, giving them stronger motivation to compete in the market with lower prices. This result shows that the market is more competitive with cross-selling strategy, and it's in accordance with intuition, given the fact that consumers have more choices on both platforms.

The more novel result is how the platform's profit changes different strategies and the commission level. If the adoption of cross-selling strategy leads to a more competitive market with lower profit, then there would be no motivation for the platforms to do that. However, our results show that it's not always the case.

In figure 3, the green line represents the equilibrium profit in the full-cross-selling equilibrium, while the purple line is the profit in the exclusive equilibrium. The other two lines represent each platform's profit when firm 1 adopts cross-selling and firm 2 doesn't. Intuitively, firm 1 will be benefited from a larger σ since they play the seller role better by selling on both platforms. On the contrary, firm 2 would appreciate a small σ, so that they can earn more from the commission.

It's also interesting that the full-cross-selling equilibrium profit is decreasing with σ. When σ is large and both firms cross-sell, the market will be more competitive with low prices and

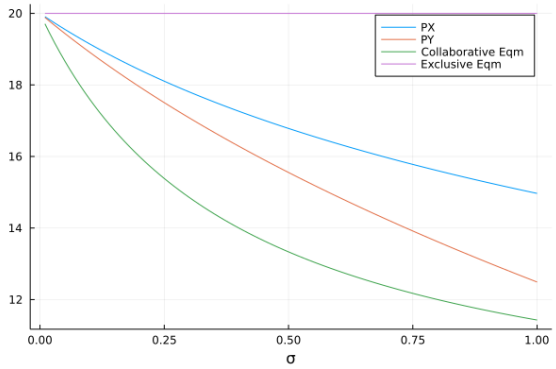


Figure 1: Price change

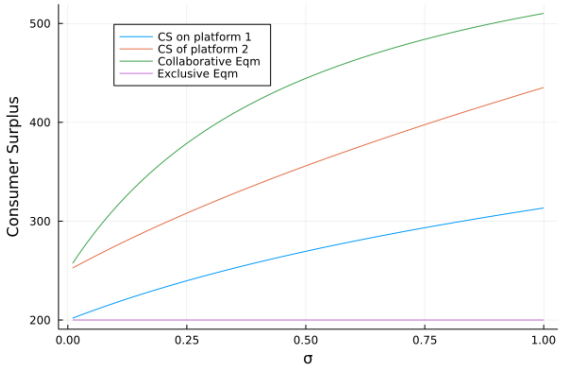


Figure 2: Welfare change

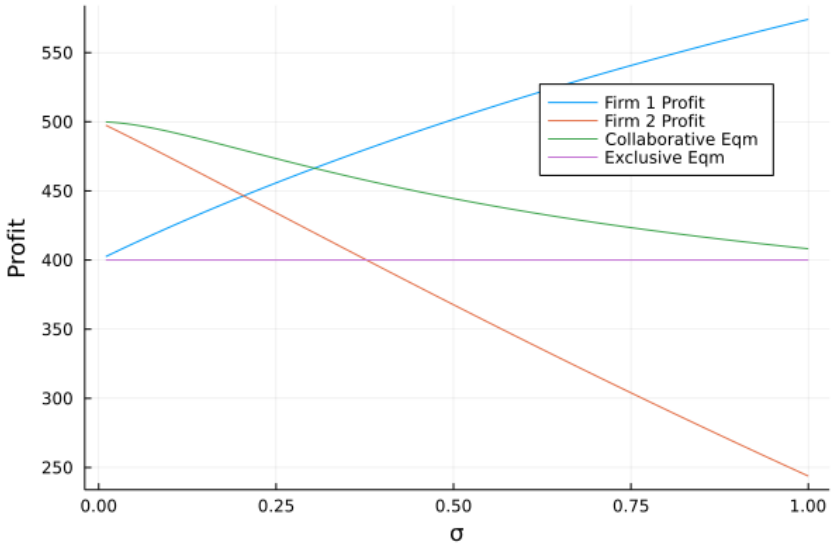


Figure 3: Profits change with commission

low profits. However, when σ goes to zero, both platforms make more profits from selling other firms' products on their platform instead of their own product on the other platform, suggesting that the marketplace role is more appreciated.

I also run a simulation with a = 40 and $\sigma = 0.6$ to observe how the output changes with the degree of product differentiation. The trends for prices and consumer surplus are pretty much the same when I change σ. A larger σ leads to lower prices and higher consumer surplus. The intuition behind is that, when the products are more alike, the market becomes more competitive which leads to low prices and high consumer surplus. What's different is how the equilibrium profits change with s. As shown in the figure, except for the exclusive equilibrium, all other equilibrium profits are decreasing with s. The full-cross-selling equilibrium profit decreases faster than the semi-cross-selling equilibrium profit.

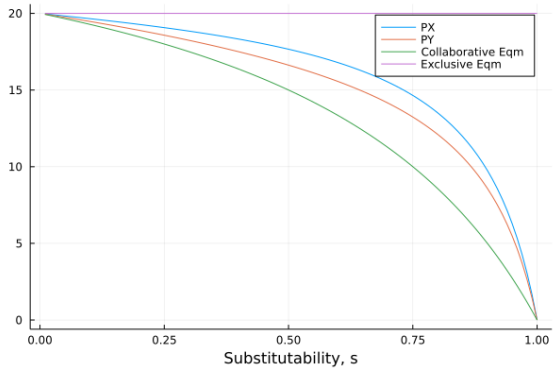


Figure 4: Price change

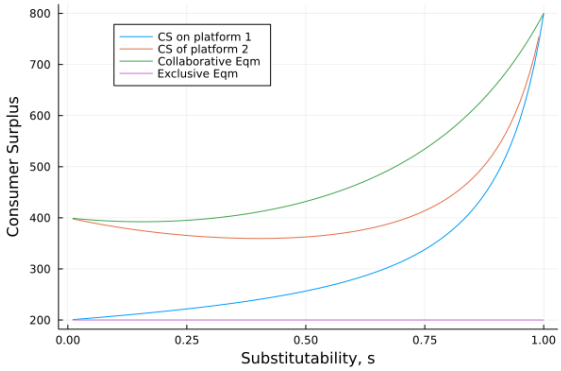


Figure 5: Welfare change

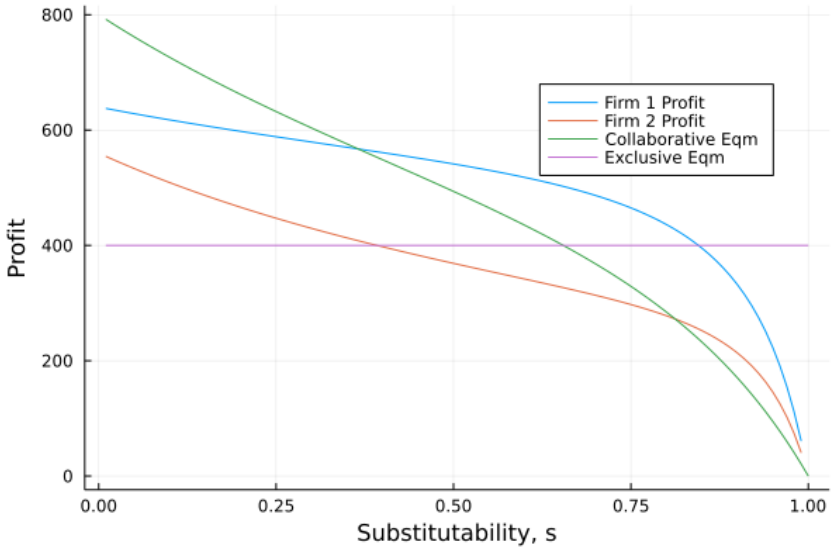


Figure 6: Profits change with substitutability

4.2 The Multihoming Case

In the multihoming case, I allow the consumers to move across platforms with a console fee c, which will be earned by the platform they moved to. In this case, even if one platform doesn't cross sells, consumers on the other platform still have access to their game. In practice, the subscription fee or the console fee of a platform is usually stable over a long period of time, and is not dramatically differentiated over platforms. Therefore, I assume the console fee in our model to be a constant.

Since consumers can move across platforms, they have access to both games. Then in all the cases, they would have the same utility functions, which can be represented by:

$$U_1 = a\left(x_1 + y_1\right) - \frac{1}{2}\left(x_1^2 + y_1^2\right) \; - \; sx_1y_1$$
$$U_2 = a\left(x_2 + y_2\right) - \frac{1}{2}\left(x_2^2 + y_2^2\right) - sx_2y_2$$

Exclusive Equilibrium

The difference among the cases lies in consumer's budget constraint and platform's profit. With no platforms cross sells, all consumers will need to pay the console fee for the access of game on the other platform. Let M1 and M2 be the consumptions of a numeraire good of the two groups of consumers, we have the budget constraints to be:

$$\begin{aligned} M_1 \; &\leq P^X x_1 + \left(P^Y + c\right) y_1 \\ M_2 \; &\leq \left(P^X + c\right) x_2 + P^Y y_2 \end{aligned}$$

Accordingly we have the demands as:

$$\begin{aligned} x_1 \; &= \frac{1}{1 \; - \; s^2} \; \left[a\left(1 - s\right) \; - \; P^X \; + \; sP^Y \; + \; sc\right] \\ y_1 \; &= \frac{1}{1 \; - \; s^2} \; \left[a\left(1 - s\right) \; - \; P^Y \; + \; sP^X \; - \; c\right] \\ x_2 \; &= \frac{1}{1 \; - \; s^2} \; \left[a\left(1 - s\right) \; - \; P^X \; + \; sP^Y \; - \; c\right] \\ y_2 \; &= \frac{1}{1 \; - \; s^2} \; \left[a\left(1 - s\right) \; - \; P^Y \; + \; sP^X \; + \; sc\right] \end{aligned}$$

When there is no cross-selling, the platforms no longer need to determine the commission $(1 - \sigma)$. They earn money from selling on both platforms with additional console fee. Their profits can be represented as the followings:

$$\Pi_1 = P^X\left(x_1 + x_2\right) + cx_2$$
$$\Pi_2 = P^Y\left(y_1 + y_2\right) + cy_1$$

At equilibrium, both platforms optimize their profit by setting the optimal price. Solving the first order conditions, we have:

$$P_E^X = P_E^Y = \frac{1 - s}{2 - s}a - \frac{c}{2} \tag{10}$$

Compared with the singlehoming exclusive equilibrium price $P_E^X = P_E^Y = \frac{a}{2}$, we can find that with positive s and c, the multihoming price will always be lower than the singlehoming case.

Note that with the console fee, it's possible to have negative demands with non-negative prices. To be more specific, to support non-negative demand y1 and x2 at equilibrium, we need:

$$c \leq \frac{2}{1+s}\frac{1-s}{2-s}a \tag{11}$$

To support a non-negative price, we need:

$$c \leq \frac{2\left(1-s\right)}{2-s}a \tag{12}$$

Since $\frac{2}{1+s} < 2 \ \forall \ s \in [0,1]$, not all non-negative price can support non-negative demands. Any negative demand or negative price is not feasible in real world, therefore, in the following analysis, I only consider the combinations of s and c such that all the demands and prices are positive.

Full-Cross-Selling Equilibrium

When both firms cross sell, there is no need for consumers to move across platforms and pay the console fee. Therefore, this case is identical with the full-cross-selling equilibrium in the singlehoming case.

Semi-Cross-Selling Equilibrium

WLOG, we assume platform 1 cross sells. Then consumers on platform 1 will need to pay the console fee for access of game y while consumers on platform 2 don't need to pay any console fee. Therefore, consumers' budgets constrains are:

$$\begin{aligned} M_1 &\leq P^X x_1 + \left(P^Y + c\right) y_1 \\ M_2 &\leq P^X x_2 + P^Y y_2 \end{aligned}$$

We can derive the demands system as follows:

$$\begin{aligned} x_1 &= a - s y_1 - P^X \\ y_1 &= a - s x_1 - P^Y - c \\ x_2 &= a - s y_2 - P^X \\ y_2 &= a - s x_2 - P^Y \end{aligned}$$

Platform 1 makes money from selling game x on both platforms, while platform 2 earns from selling game y on both platforms, selling game x on platform 2 and the console fee. Their profits can be written as:

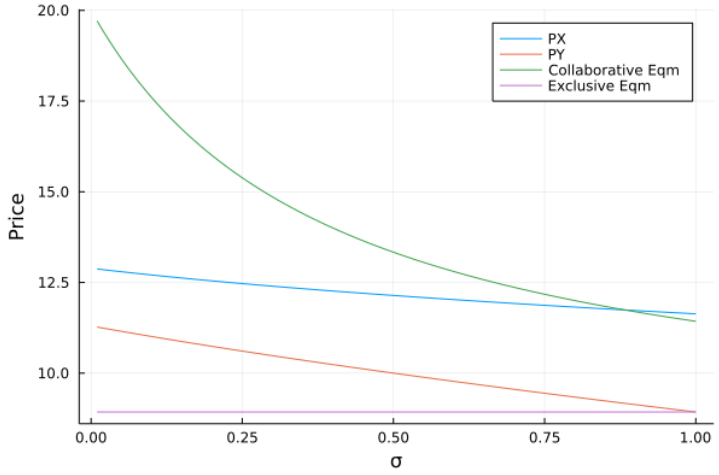


Figure 7: Price change with commission

$$\Pi_1 = P^X (x_1 + \sigma x_2)$$
$$\Pi_2 = P^Y (y_2 + \sigma y_1) + (1 - \sigma) P^X x_2 + c y_2$$

Solving the first order condition, we can find the analytical solutions for px and py. Since they are complicated in notation, I will leave the detailed deriving process and solution in appendix.

Equilibrium Comparison

There are many similar trends in the multihoming case and the singlehoming case. The equilibrium prices in semi-cross-selling equilibrium and full-cross-selling equilibrium decrease with σ while the consumer surplus increases. One noticeable difference is that the exclusive equilibrium now has the lowest price and highest consumer surplus, while the full-cross-selling case has the highest price and lowest consumer surplus.

To make it more clear, I draw a 3D graph of the difference in consumer surplus between the two equilibriums(Exclusive eqm – Full-cross-sale eqm). The parameters interested are σ and the console fee since they essentially differentiated the two equilibriums. We can see that the difference in consumer surplus is larger with a smaller σ and a larger console fee.

This result may seem counterintuitive because normally economists believe an exclusive market has higher prices and thus lower consumer surplus. I will discuss in general when would this happen in sections 7. A brief intuitve explanation for this finding is collusion. In an exclusive market, the platforms don't interact with each other until the price competition, while in the full-cross-selling market, they will bargain over the commission, and take that into

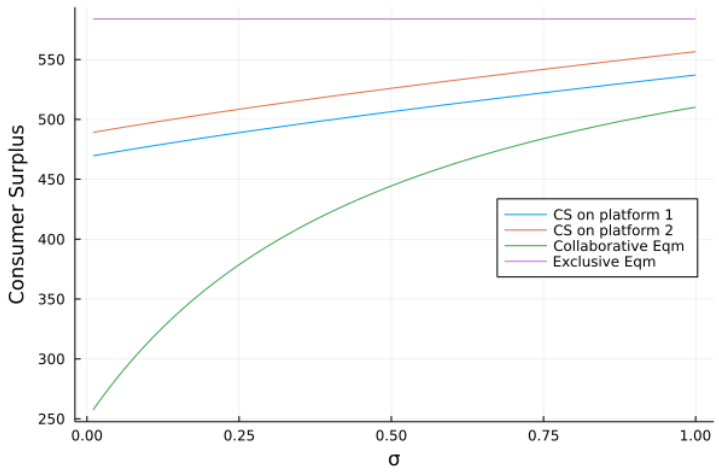


Figure 8: Consumer Surplus Change with Commission

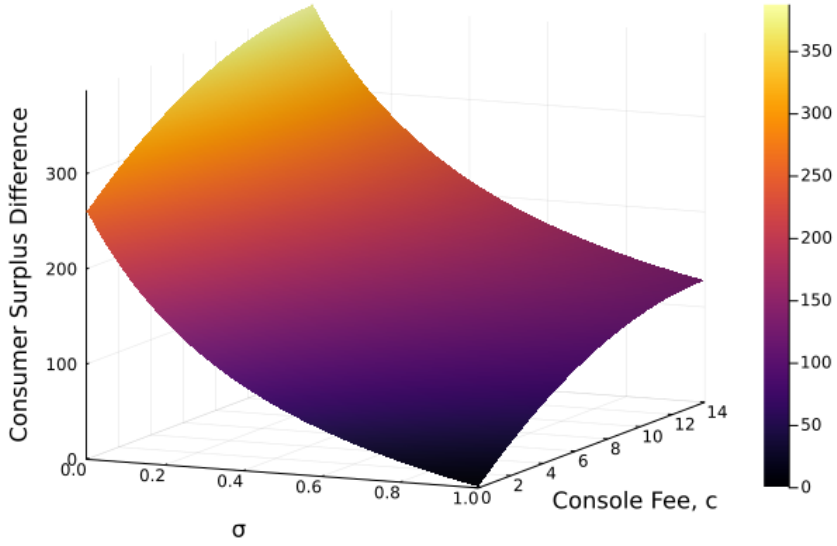


Figure 9: Consumer Surplus Difference

the price competition. This commission gives them an incentive and the same time a tool to collude so that they together drive up the market price and make more profit.

Similar trends are also shared in equilibrium profits between the multihoming case and the singlehoming case. Firm 1 is benefited from a large σ while firm 2 prefers a large commission. The key difference is that the exclusive equilibrium now has the all-time lowest profit. At the same time, all the equilibrium profits are decreasing with substitutability which is due to the more intensive competition in the market.

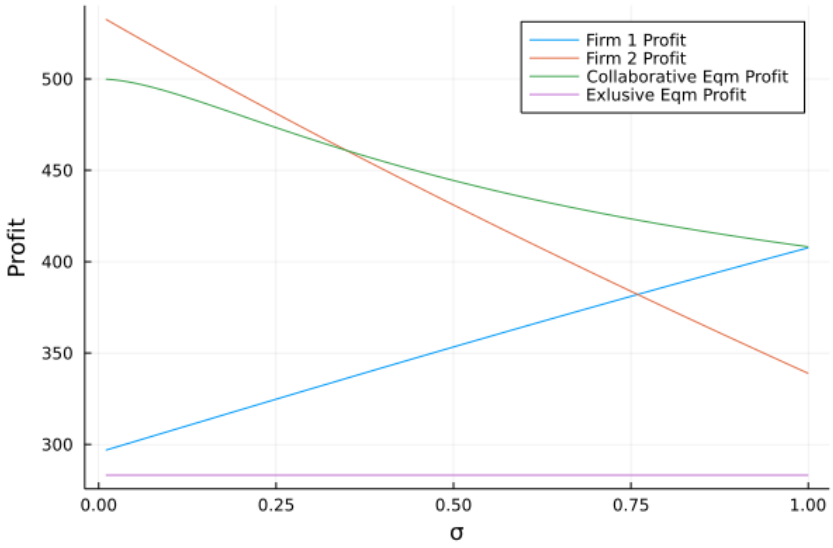


Figure 10: Profit change with commission

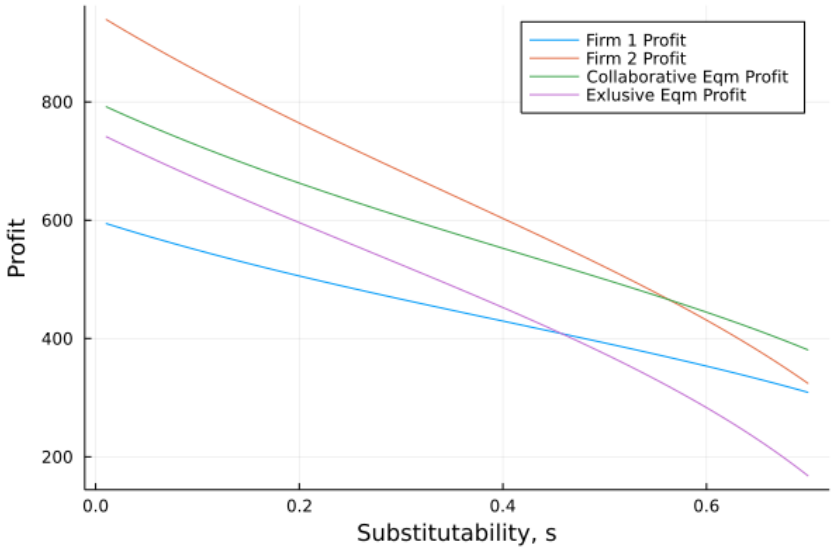


Figure 11: Profit change with substitutability

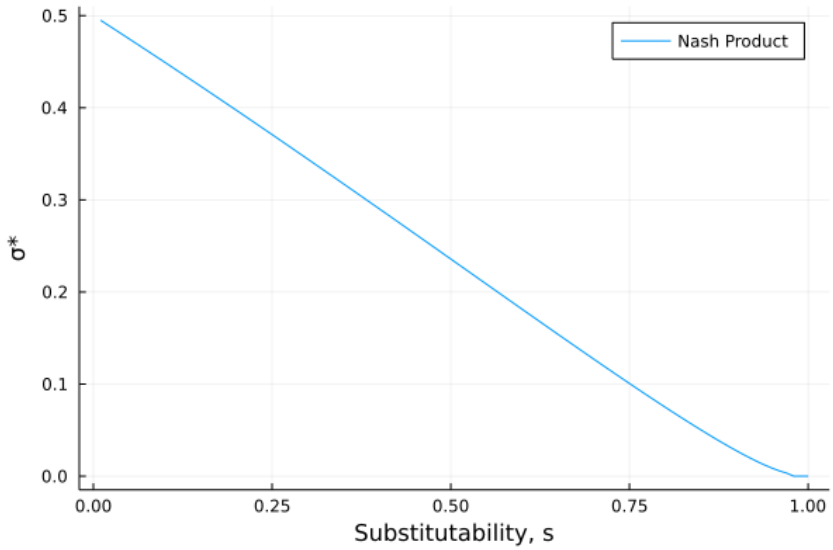


Figure 12: Negotiated σ under singlehoming

5 Stage 2: Nash Bargaining

If there is no need for bargaining in an exclusive equilibrium since the output is constant with σ. In a full-cross-selling equilibrium, both platforms would agree to have $\sigma = 0$ to maximize the equilibrium profit. This is an interesting contracting mode that's different from Hagiu and LeeHagiu and Lee (2011)'s "outright sale", when the platform obtains both control over pricing and all the revenue from sale, and "affiliation", when content provider sets their price and keeps revenue from sale. The content provider, in this case the other platform, set the price for the content, while the selling platform receives all the revenue form sales.

The firms only have incentives to bargain when only one firm wish to sell their superstar product on the other platform. A disagreement of import will lead to no cross-selling, thus the disagreement payoff for both firms are exclusive equilibrium profit. Following NashNash (1953), we denote the Nash product $B(\sigma)$ as the follows:

$$B(\sigma) = (\Pi_1(\sigma) - \Pi_{NC})(\Pi_2(\sigma) - \Pi_{NC}) \tag{13}$$

Where $\Pi_1(\sigma)$ and $\Pi_2(\sigma)$ are the equilibrium profits for platform 1 and 2 respectively.

For the semi-cross-selling case, $B(\sigma)$ is strictly concave in $\sigma \in (0, 1)$, therefore, we can always find $\sigma^S(s)$ for the singlehoming case and $\sigma^M(s, c)$ for the multihoming case that maximize the Nash product respectively. Figure illustrates the negotiated σ using parameter values $a = 40$. We can see that for the singlehoming case, the $\sigma^S(s)$ is always decreasing with substitutability towards 0. For the multihoming case, I have a 3D figure for $\sigma^M(s, c)$. As shown in the figure,$\sigma^M(s, c)$ is decreasing with both substitutability and the console fee.

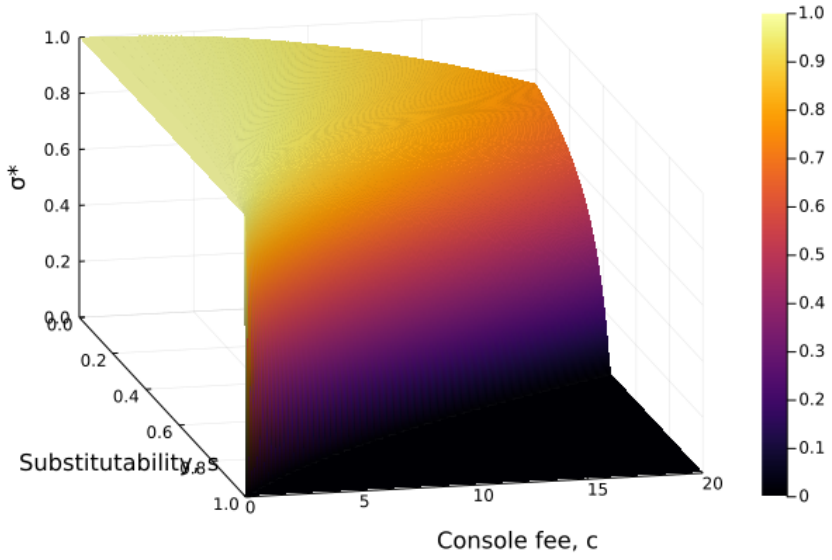


Figure 13: Negotiated σ under multihoming

6 Stage 1: Strategy Selection

With the result of Stage 2 in hand, let us now move back to stage 1 and determine the outcome of the strategy selection game. I set $a = 40$ in the following simulation.

I denote the payoffs of the first-stage game with table6. Both firm at stage 1 have 2 strategies: sell exclusively or cross sell. Their payoffs are determined by the bertrand competition at the negotiated commission rate. Π^E represents the exclusive equilibrium profit, while Π^C is the full-cross-selling equilibrium profit. Π^1 and Π^2 are the equilibrium profit for firm 1 and firm 2 respectively in a semi-cross-selling equilibrium.

	Exclusive	Cross Sell
Exclusive	(Π^E, Π^E)	(Π^2, Π^1)
Cross Sell	(Π^1, Π^2)	(Π^C, Π^C)

Table 1: Payoff at Stage 1

6.1 Singlehoming case

Figure 14 illustrate how the equilibrium profits change with the substitutability. In the singlehoming case, the equilibrium profit of firm 1 decreases when they are the only firm cross-selling or when both firm cross-sells, while the exclusive equilibrium profit doesn't change with substitutability. The differences among the equilibrium profits shrink as the substitutability increases, suggesting that when the products are more alike, the platform earns less in equilibrium due to a higher level of market competition.

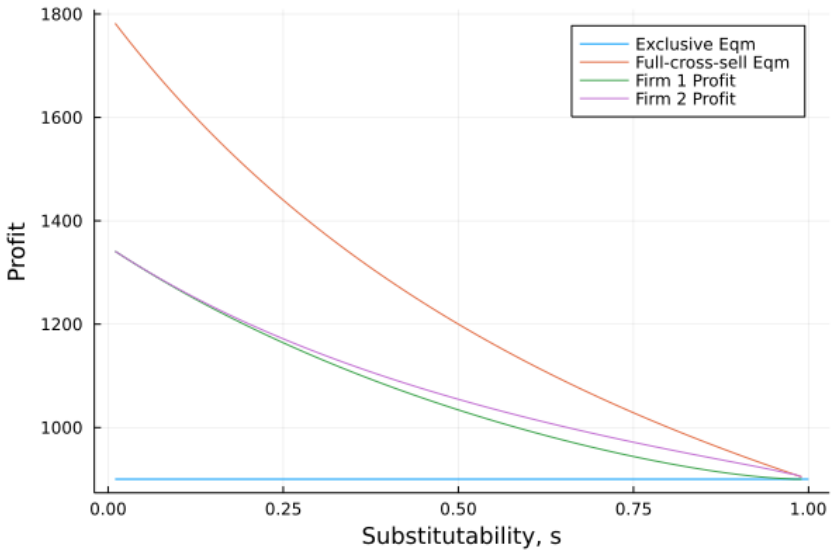


Figure 14: Equilibrium profits under singlehoming

As we can see in figure 14, with any substitutability, $\Pi^E < \Pi^1 < \Pi^2 < \Pi^C$, so that both firms would adopt the cross-selling strategy.

6.1.1 Multihoming Case

The following figures gives a snapshot of how the equilibrium profits of the two firms change with substitutability at a fixed console fee($c = 15$). Quite similar to the singlehoming case that all the equilibrium profits are decreasing with s. The difference is that now exclusive equilibrium profit is also decreasing with s due to the decrease of $\sigma*$.

I also draw 3D figures for firm 1 and 2's profit in a semi-cross-selling equilibrium. Overall, the two firms' equilibirum profits behave similar as they decrease with substitutability and increase with the console fee. As shown in the figure, firm 1's profit decreases with substitutability and console fee, while firm 2's profit is increasing with console fee. Firm 2's profit has a quadratic relationship with substitutability as the equilibrium profit shows a slight upward trend when s is very close to its upper limit. We make the profits zero when the combination of s and c leads to negative demand, which is the reason for the cliffs shown in the profit figures.

To better compare the equilibirum profits, I draw a heatmap of the difference betweem the exclusive equilibirum and the full-cross-selling equilibrium with respect to the substitutability and the console fee. As showed in figure 18, the difference is always positive, and is increasing as the substitutability and the console fee.

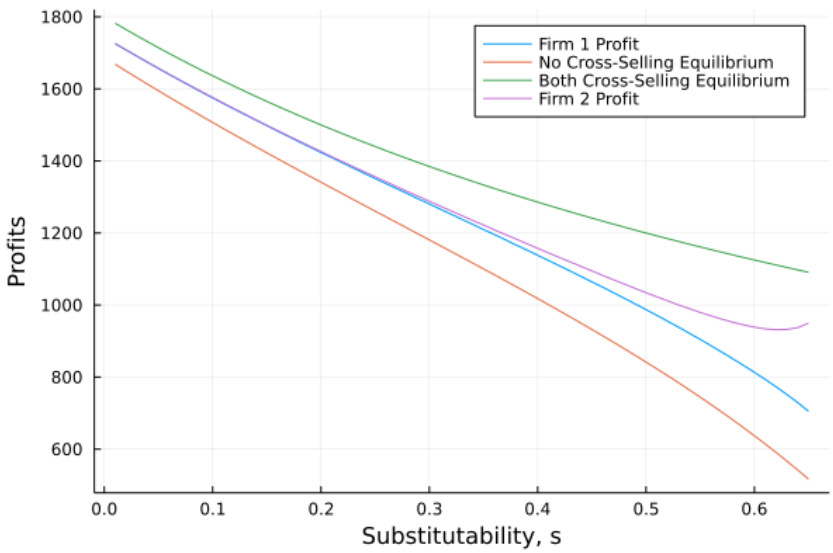


Figure 15: Equilibrium profits under multihoming

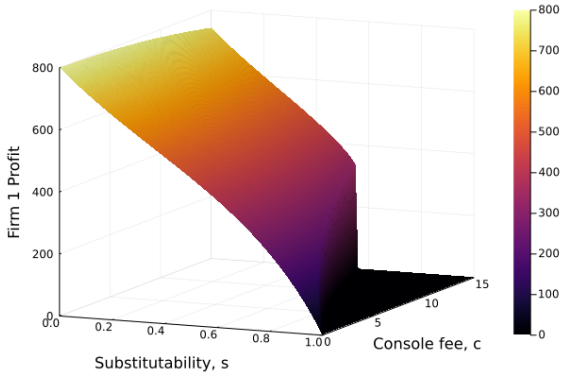


Figure 16: Firm 1 Equilibirum Profit

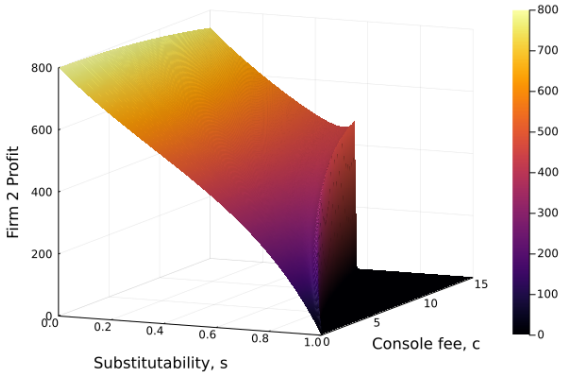


Figure 17: Firm 2 Equilibirum Profit

7 General Form Analysis

In this sector, I take a closer look at why consumer surplus is lower in the full-cross-selling equilibrium with a general form of demand system. In our game, firms optimize their profit, so the equilibrium price will always be higher than the consumer surplus optimizing price and a higher equilibrium price will lead to a lower consumer surplus and vice versa. Given this, I simply the problem to examine how the equilibrium prices change with σ and c.

I generalized our model as follows. Let P^X and P^Y denote the price of product X and product Y, respectively. I assume the consumers on both platforms have the same preference, and their demands for product X and product Y can be represented by $x\left(P^X, P^Y\right)$ and $y\left(P^X, P^Y\right)$.

First, I want to show the sufficient condition to have the an unique symmetrical equilibrium.

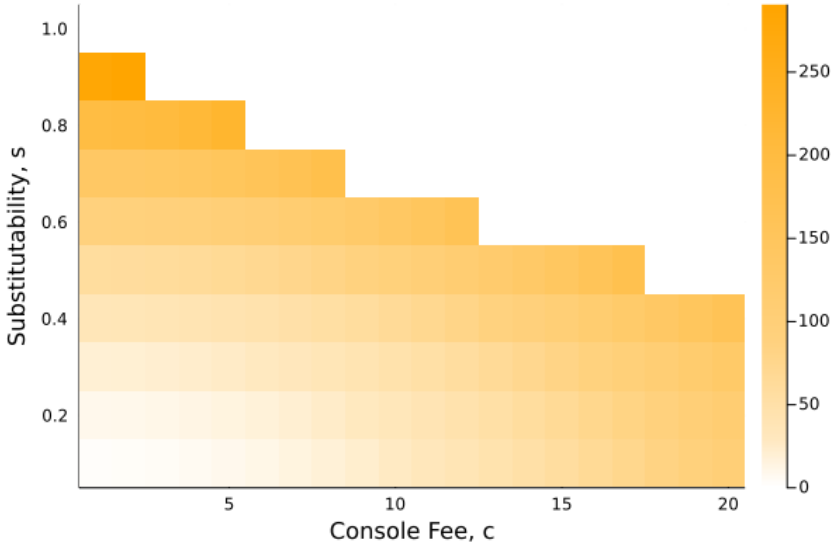


Figure 18: Difference between exclusive and full-cross-selling equilibrium

Assumption 1. *The demands for the products are symmetrical, i.e.,*

$$x\left(P^X, P^Y\right) \;=\; y\left(P^Y, P^X\right) \tag{14}$$

Assumption 2. *i).The demand for a product is decreasing with its own price but increasing with the rival's price, i.e.,*

$$\frac{\partial x}{\partial P^X} \;<\; 0, \;\; \frac{\partial x}{\partial P^Y} \;>\; 0 \tag{15}$$

ii). The demand is concave in its own price and its rival's price, i.e.,

$$\frac{\partial^2 x}{\partial {P^X}^2}, \;\; \frac{\partial^2 x}{\partial {P^Y}^2} \leq 0 \tag{16}$$

iii). The cross second derivative is non-negative, i.e.,

$$\frac{\partial^2 x}{\partial P^X P^Y}, \;\; \frac{\partial^2 y}{\partial P^X P^Y} \geq 0 \tag{17}$$

The profits of both platforms in a full-cross-selling equilibrium can be written as:

$$\Pi_1 = (1+\sigma)\, P^X x\left(P^X, P^Y\right) + (1-\sigma)\, P^Y y\left(P^X, P^Y\right)$$
$$\Pi_2 = (1-\sigma)\, P^X x\left(P^X, P^Y\right) + (1\;+\;\sigma)\, P^Y y\left(P^X, P^Y\right)$$

The the FOCs are:

$$\frac{\partial \Pi_1}{P^X} = (1+\sigma)\left(P^X \frac{\partial x}{\partial P^X}\left(P^X, P^Y\right) + x\left(P^X, P^Y\right)\right) + (1-\sigma)\, P^Y \frac{\partial y}{\partial P^X}\left(P^X, P^Y\right)$$

$$\frac{\partial \Pi_2}{P^Y} = (1+\sigma)\left(P^Y \frac{\partial y}{\partial P^Y}\left(P^X, P^Y\right) + y\left(P^X, P^Y\right)\right) + (1-\sigma)\, P^X \frac{\partial x}{\partial P^Y}\left(P^X, P^Y\right)$$

The firms would optimize their profits by selecting their own price and this process implicitly defined their best-response.

$$G(P^X, P^Y) = \frac{\partial \Pi_1}{P^X} = 0$$
$$H(P^X, P^Y) = \frac{\partial \Pi_2}{P^Y} = 0$$

Assumption 3. *i). The demand of a product react more dramatically to its own price than its rival's price at equilibrium, i.e.,*

$$\left|\frac{\partial x}{\partial P^X}\right| \geq \left|\frac{\partial x}{\partial P^Y}\right|, \ \forall (P^X, P^Y) \ s.t. \ G(P^X, P^Y) = 0 \tag{18}$$

ii). The demand reaction of one product to its own price is more dramatic than the average of how the two products' demands react to their rivals' prices at equilibrium.

$$\left|\frac{\partial x}{\partial P^X}\right| \geq \frac{1}{2}\left|\left(\frac{\partial x}{\partial P^Y} + \frac{\partial y}{\partial P^X}\right)\right|, \ \forall (P^X, P^Y) \ s.t. \ G(P^X, P^Y) = 0 \tag{19}$$

iii). The marginal change of a product's demand to its own price react more to its own price than it's rival's price, i.e.,

$$|\frac{\partial^2 x}{\partial {P^X}^2}| \geq |\frac{\partial^2 x}{\partial P^X P^Y}|, \ \forall (P^X, P^Y) \ s.t. \ G(P^X, P^Y) = 0 \tag{20}$$

Proposition 1. *Given **Assumption 1-3**, there exists an unique symmetrical equilibrium with price P^**

Proof. By Implicit Function Theorem, firm 1's best-response function $P^X = BR_1(P^Y)$ is implicitly defined by $G(P^X, P^Y) = 0$. Therefore, we can derive its slope as:

$$\frac{\partial P^Y}{\partial P^X} = -\frac{\partial G}{\partial P^X} / \frac{\partial G}{\partial P^Y}$$

where

$$\frac{\partial G}{\partial P^X} = 2(1+\sigma)\frac{\partial x}{\partial P^X} + (1+\sigma)P^X \frac{\partial^2 x}{\partial {P^X}^2} + (1-\sigma)P^Y \frac{\partial^2 y}{\partial {P^Y}^2} < 0$$

$$\frac{\partial G}{\partial P^Y} = (1+\sigma)\frac{\partial x}{\partial P^Y} + (1-\sigma)\frac{\partial y}{\partial P^X} + (1+\sigma)P^X \frac{\partial^2 x}{\partial P^X P^Y} + (1-\sigma)P^Y \frac{\partial^2 y}{\partial P^Y P^X} > 0$$

Therefore, if at some point (P_1^X, P_1^Y)

$$\frac{\partial x}{\partial P^Y}(P_1^X, P_1^Y) \geq \frac{\partial y}{\partial P^X}(P^X, P^Y)$$

Then by **Assumption 3.1**:

$$2(1+\sigma)|\frac{\partial x}{\partial P^X}| \geq 2(1+\sigma)|\frac{\partial x}{\partial P^Y}| > (1+\sigma)\frac{\partial x}{\partial P^Y} + (1-\sigma)\frac{\partial y}{\partial P^X}$$

Otherwise, we have

$$\frac{\partial x}{\partial P^Y}(P_1^X, P_1^Y) < \frac{\partial y}{\partial P^X}(P_1^X, P_1^Y)$$

Then by **Assumption 3.2**:

$$\begin{aligned} &2(1+\sigma)|\frac{\partial x}{\partial P^X}| > 2|\frac{\partial x}{\partial P^X}| \\ &> \frac{\partial x}{\partial P^Y} + \frac{\partial y}{\partial P^X} > (1+\sigma)\frac{\partial x}{\partial P^Y} + (1-\sigma)\frac{\partial y}{\partial P^X} \end{aligned}$$

Furthermore, by **Assumption 3.3**, we have:

$$-\frac{\partial G}{\partial P^X} \geq \frac{\partial G}{\partial P^Y} \Rightarrow \frac{\partial P^Y}{\partial P^X} \geq 1$$

In the same manner, firm 2's best-response $P^Y = BR_2(P^X)$ is implicitly defined by $H(P^X, P^Y) = 0$, and we have

$$\frac{\partial P^Y}{\partial P^X} = -\frac{\partial H}{\partial P^X} / \frac{\partial H}{\partial P^Y} < 1$$

Therefore, the two best-respond functions can have at most 1 crossing, suggesting that there can be at most 1 equilibrum.

When $P^Y \to 0$, the FOC of firm 1 is transformed into

$$(1+\sigma)\left(P^X \frac{\partial x}{\partial P^X}\left(P^X, 0\right) + x\left(P^X, 0\right)\right) = 0$$

Since $x\left(P^X, 0\right) > 0$ and $P^X \frac{\partial x}{\partial P^X}\left(P^X, 0\right) < 0$, we can find a positive x that satisfies the FOC. Therefore, the best-respond function of firm 1 must have a crossing with the line $P^Y = P^X$. By the symmetry assumed in **Assumption 1**, for any given σ the two best-respond line must have a crossing when $P^Y = P^X = P^*$ for some P^X.

□

Secondly, I want to show what conditions are required to have the full-cross-selling equilibrium price decrease with σ.

Let $f(P^X, P^Y) = P^X x(P^X, P^Y)$ denote a firm's revenue from their own product.

Assumption 4: At equilibrium, $\frac{\partial f}{\partial P^X}(P^*, P^*)$ is decreasing with the equilibrium price while $\frac{\partial f}{\partial P^Y}(P^*, P^*)$ is increasing with the equilibrium price.

This assumption is a extension of **Assumption 2** as it further constrains the realationships between the first and second derivatives of the demand function. As for economic intuition, this

assumption essentially guarantees that as the equilibrium price increases, the marginal cost of increasing my own price increases and the marginal benefit of increasing my rival's price also increases.

Proposition 2. *With **Assumption 1-4**, the equilibirum price P^* decreases with σ*

Proof. Given

$$\frac{\partial x}{\partial P^Y} > 0$$

and

$$(1+\sigma)\left(P^* \frac{\partial x}{\partial P^X}(P^*, P^*) + x(P^*, P^*)\right) + (1-\sigma) P^* \frac{\partial x}{\partial P^Y}(P^*, P^*) = 0$$

We must have

$$P^* \frac{\partial x}{\partial P^Y}(P^*, P^*) > 0$$

And

$$P^* \frac{\partial x}{\partial P^X}(P^*, P^*) + x(P^*, P^*) < 0$$

We can solve for σ:

$$\sigma = \frac{P^* \frac{\partial x}{\partial P^Y}(P^*, P^*) + \left(P^* \frac{\partial x}{\partial P^X}(P^*, P^*) + x(P^*, P^*)\right)}{P^* \frac{\partial x}{\partial P^Y}(P^*, P^*) - \left(P^* \frac{\partial x}{\partial P^X}(P^*, P^*) + x(P^*, P^*)\right)}$$

Since

$$P^* \frac{\partial x}{\partial P^Y}(P^*, P^*) = \frac{\partial f}{\partial P^Y}\left(P^X = P^*, P^Y = P^*\right)$$
$$P^* \frac{\partial x}{\partial P^X}(P^*, P^*) + x(P^*, P^*) = \frac{\partial f}{\partial P^X}\left(P^X = P^*, P^Y = P^*\right)$$

Then $\frac{\partial f}{\partial P^Y}(P, P)$ is increasing with P and $\frac{\partial f}{\partial P^X}(P, P)$ is decreasing with P guarantees that σ is decreasing with P* and vice versa.

Since the firms would like to have a higher equilibirum price for more profit, this result suggest that they have incentives to negotiate and set $\sigma = 0$.

□

Thirdly, I want to show that the equilibrium price decreases with the console fee c.

The equilibrium profits for both platforms in an exclusive equilibrium can be written as:

$$\Pi_1 = P^X(x_1 + x_2) + cx_2$$
$$\Pi_2 = P^Y(y_1 + y_2) + cy_1$$

Where x1 and x2 are the demand for X on both platforms, and y1 and y2 are the demands for Y on both platforms.

Since the console fee is the only difference on the demand side between both platforms, we have

$$x_1\left(P^X, P^Y\right) = x\left(P^X, P^Y\right)$$
$$x_2\left(P^X, P^Y\right) = x\left(P^X + c, P^Y\right)$$

Where x is the general form demand function I used in previous analysis.

Proposition 3. *With **Assumption 1-4**, the exclusive equilibrium price is always lower than the full-cross-selling equilibrium price.*

Proof. When $\sigma = 1$, at equilibrium we have

$$2\left(P^* \frac{\partial x}{\partial P^X}\left(P^*, P^*\right) + x\left(P^*, P^*\right)\right) \;=\; 0$$

With **Assumption 1-4**, there exist an exclusive equilibrium with price P^E, then we have the FOC:

$$x\left(P^E, P^E\right) + x\left(P^E + c, P^E\right) + P^E \frac{\partial x}{\partial P^X}\left(P^E, P^E\right) + \left(P^E + c\right) \frac{\partial x}{\partial P^X}\left(P^E + c, P^E\right) \;=\; 0$$

By **Assumption 2**, with a non-negative c,

$$x\left(P^E + c, P^E\right) \leq x\left(P^E, P^E\right),\; P^E \frac{\partial x}{\partial P^X}\left(P^E + c, P^E\right) \leq P^E \frac{\partial x}{\partial P^X}\left(P^E, P^E\right)$$

Therefore, if $P^* = P^E$, the FOC of the exclusive equilibirum should be smaller than 0.

By **Assumption 4**, $x\left(P, P\right) + P \frac{\partial x}{\partial P^X}\left(P, P\right)$ decrease with P, thus, $P^E < P^*$. □

8 Conclusion

Vertical integration and exclusivity are both commonly observed in two-sided markets. This paper studies platforms' exclusive strategy rationale and its competitive effects when they are vertically integrated with content providers. I found that the platforms indeed have the motivation to give up their exclusivity and supply their rival in a full-cross-selling equilibirum. Compared with a semi-cross-selling equilibrium and an exclusive equilibirum, both firms achieve more profit to cross sell. This result holds true no matter the consumers are singlehoming or multihoming.

It's also interesting that the full-cross-selling equilibrium profit is increasing with the commission so both platforms would wish to have the commission be 100% of the sales revenue. This leads to a new contracting mode that the content provider has full control over content

pricing while the selling platform keeps all the sales revenue. In other words, the seller would let his/her rival "take the wheel" over the pricing of part of their goods to maximize profit.

As for the competitive effect, the full-cross-selling equilibrium generates a consumer surplus higher than the singlehoming exclusive equilibrium but lower than the multihoming exclusive equilibrium. One potential explanation is that the exclusive equilibrium under singlehoming works as a monopoly market while it works as a competitive market under multihoming. The bargaining over commission gives the two platforms a tool to collude so that they can charge a higher price on the whole market and drives consumer surplus down.

Though our paper focuses primarily on the video game industry, testable implications are also provided in other sectors and industries. In the online video industry, many platforms are invested in first-party titles and vertical mergers, and they can choose between streaming exclusively or selling to their rivals based on a commission per transaction mode. Our analysis conjectures that exclusivity would make the market price lower, especially in a fiercely competed market. In a similar way, our results also provide insights into online shopping platforms. This market features both vertical integration and numerical sale. Our model predicts that the platforms can make more profits by giving up exclusivity of their first-party products.

From a policy perspective, our results provide a new lens to understand the competitive effect of exclusivity. When platforms are vertically integrated with content providers, their dual roles make their behavior complicated. Our results suggest that exclusivity does not necessarily favor the firms. Rather, under some conditions, firms prefer to give up exclusivity and share the market with a negotiated commission.

Our paper also brings new understands of two-sided platform, vertical integration and collusion. While rich literature has studies how vertical integrations can bring upstream firms or downstream firms ability to collude, yet few focus on collusion in a two-sided market. Our results show that vertical integrations indeed generate incentives for platforms to collude using commission as a media, and can potentially have anti-competitive effect.

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Appendix A Solution of Semi-cross-selling Equilibrium

Singlehoming Case

W.L.O.G, let's call the firm who cross sells firm 1, and the other one firm 2. Since consumers are singlehoming, they only have access to product 1 if they are on platform 1, while they have access to both products if they are on platform 2. Therefore, we can formalize the optimization problem of the two groups of consumers as follows:

$$\max_{x_1} U_1(x_1) - P^X x_1$$
$$\max_{x_2,y_2} U_2(x_2, y_2) - P^X x_2 - P^Y y_2$$

After solving the FOC, we can derive their demands as ;

$$\begin{aligned} x_1 &= a - P^X \\ x_2 &= \frac{1}{1-s^2}\left[a\,(1-s) - P^X + sP^Y\right] \\ y_2 &= \frac{1}{1-s^2}\left[a\,(1-s) - P^Y + sP^X\right] \end{aligned}$$

Then we can implement the demands into firms' profit functions

$$\begin{aligned} \Pi_1 &= P^X x_1 + \sigma P^X x_2 \\ &= P^X(a - P^X) + \sigma P^X \frac{1}{1-s^2}\left[a\,(1-s) - P^X + sP^Y\right] \end{aligned}$$

and

$$\begin{aligned} \Pi_2 &= P^Y y_2 + (1-\sigma)P^X x_2 \\ &= P^Y \frac{1}{1-s^2}\left[a\,(1-s) - P^Y + sP^X\right] + (1-\sigma)P^X \frac{1}{1-s^2}\left[a\,(1-s) - P^X + sP^Y\right] \end{aligned}$$

Firm 1 can only control P^X and firm 2 can only control P^Y, so their FOC can be derived as :

$$\begin{aligned} \frac{\partial \Pi_1}{P^X} = 0 &\Rightarrow a(1-s)(1+s+\sigma) + s\sigma P^Y = 2(1-s^2) + \sigma P^X \\ \frac{\partial \Pi_2}{P^Y} = 0 &\Rightarrow P^Y = \frac{1}{2}\left[a(1-s) + (2-\sigma)sP^X\right] \end{aligned}$$

We can solve P^X and P^Y by implementing them to each other. The solutions are as follows:

$$P^X = \frac{a(1-s)(2+2s+2\sigma+s\sigma)}{[4(1-s^2+\sigma)-s^2\sigma(2-\sigma)]}$$

$$\begin{aligned} P^Y &= \frac{1}{2}a(1-s) + \frac{(2-\sigma)s}{2}P^X \\ &= \frac{1}{2}a(1-s) + \frac{(2-\sigma)s}{2}\frac{a(1-s)(2+2s+2\sigma+s\sigma)}{[4(1-s^2+\sigma)-s^2\sigma(2-\sigma)]} \end{aligned}$$

The numerator of P^X is obvious positive. Its denominator is increasing in σ and decreasing in s, and it's always positive except the extreme case where $\sigma = 0, s = 1$ and the denominator goes to 0. P^Y is therefore also positive.

Moreover, it can be shown that

$$P^Y - P^X = \frac{-2s\sigma(s+\sigma)a(1-s)}{s\,[4(1-s^2+\sigma)-s^2\sigma(2-\sigma)]} \leq 0$$

Multihoming Case

Similar as the previous case, we can use consumers' optimazation problem to derive the demand functions.

$$\begin{aligned} x_1 &= \frac{1}{1-s^2}\left[a(1-s) + sP^Y + sc - P^X\right] \\ x_2 &= \frac{1}{1-s^2}\left[a(1-s) + sP^Y - P^X\right] \\ y_1 &= \frac{1}{1-s^2}\left[a(1-s) + sP^X - c - P^Y\right] \\ y_2 &= \frac{1}{1-s^2}\left[a(1-s) + sP^X - P^Y\right] \end{aligned}$$

Implementing the demands into firms' profit functions, we have:

$$\begin{aligned} \Pi_1 &= P^X x_1 + \sigma P^X x_2 \\ &= P^X \frac{1}{1-s^2}\left[a(1-s) + sP^Y + sc - P^X\right] + \sigma P^X \frac{1}{1-s^2}\left[a(1-s) + sP^Y - P^X\right] \end{aligned}$$

$$
\begin{aligned}
\Pi_2 &= (1-\sigma)P^X x_2 + P^Y(y_1+y_2) + cy_1 \\
&= (1-\sigma)P^X \frac{1}{1-s^2}\left[a(1-s) + sP^Y - P^X\right] + P^Y \frac{1}{1-s^2}\left[2a(1-s) + 2sP^X - 2P^Y - c\right] \\
&+ c\frac{1}{1-s^2}\left[a(1-s) + sP^X - c - P^Y\right]
\end{aligned}
$$

Then we can solve P^X and P^Y by the FOCs:

$$
\begin{aligned}
P^X &= \frac{1}{2}\left[a(1-s) + \frac{sc}{1+\sigma} + sP^Y\right] \\
P^Y &= \frac{1}{8-s^2(3-\sigma)}\left\{(1-s)[as(3-\sigma)+4a] - c(4 - \frac{3-\sigma}{1+\sigma}s^2)\right\}
\end{aligned}
$$